

9 The human genome contains sequences of DNA called endogenous retroviruses (ERVs).

These ERVs result from infection with retroviruses.

Retroviruses are a group of RNA viruses that have a replication cycle similar to that of human immunodeficiency virus (HIV).

This replication cycle results in latency.

Some of the ERVs in the human genome have been present for millions of years, because the ERVs are passed down through many generations.

- *(a) Explain how retroviruses introduce ERVs into human cells and how the ancient ERVs can be transmitted to offspring in families.

(6)

[illegible]

(b) One group of ERVs code for viral envelope proteins.

In normal cells, these proteins are not synthesised due to epigenetic regulation.

Explain how epigenetic regulation prevents the synthesis of these viral envelope proteins.

(3)

(c) In cancer cells, the viral envelope proteins are synthesised and incorporated into the cancer cell membrane.

Explain why these viral envelope proteins could be used in a vaccine against cancer cells.

Use the information in the question to support your answer.

(5)

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(Total for Question 9 = 14 marks)

TOTAL FOR PAPER = 90 MARKS